



MKTG6205A · DOCTORAL SEMINAR

Contemporary Topics in Behavioural Marketing

Consumer-AI interactions and emerging methods

Prof. Christian Hildebrand

University of St.Gallen (HSG) · Visiting Professor, CUHK Department of Marketing
Term 1, 2026–27 · Fridays 14:30–17:15 · CYT 1128A



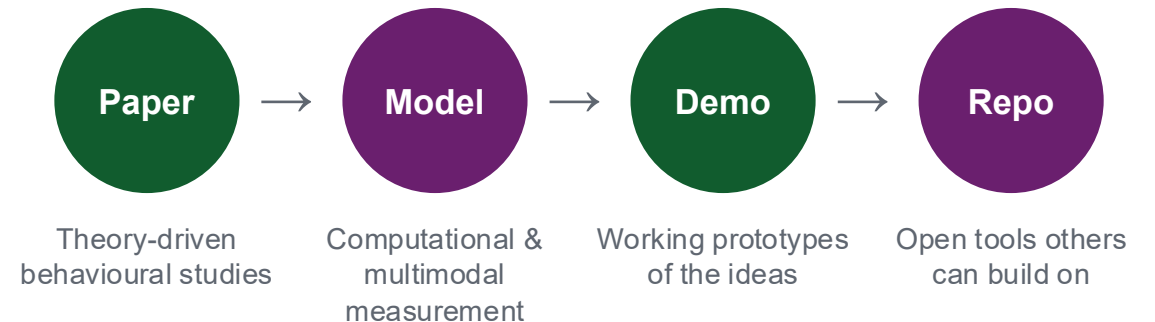
My Own Background & Academic Journey

Christian Hildebrand

- Full Professor of Marketing Analytics, University of St.Gallen (HSG)
- Executive Director, Institute of Behavioral Science & Technology (IBT-HSG)
- Visiting Professor, CUHK Department of Marketing (2026–2029)

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Research focus



Consumer-AI interaction · voice and conversational AI · multimodal behavioural data · AI and decision making

My Research Program at the “AI Impact Lab”



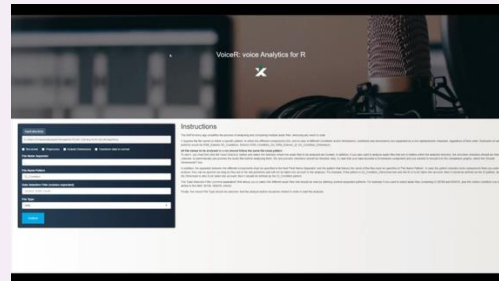
Consumer – AI Experiences (ConsumerX)

Machine Talk, JCR (2023)
Consumer Service Bots, JCR (2023)



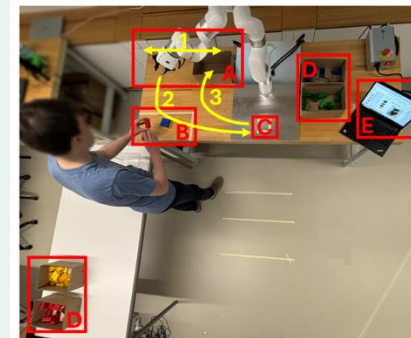
Language & Voice Analytics (VoiceX)

Audio Recording Biases, BRM (2023)
voiceR, CRAN (2022)



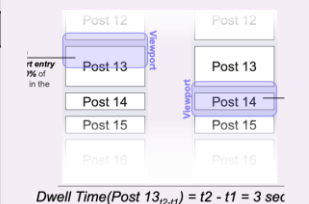
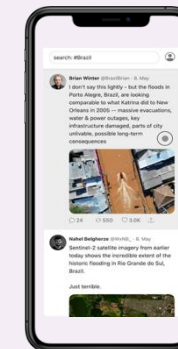
Interaction Design & Embodied AI (DesignX)

Adaptive Robots, CHI (2025)
Movement Patterns, IEEE (2022)

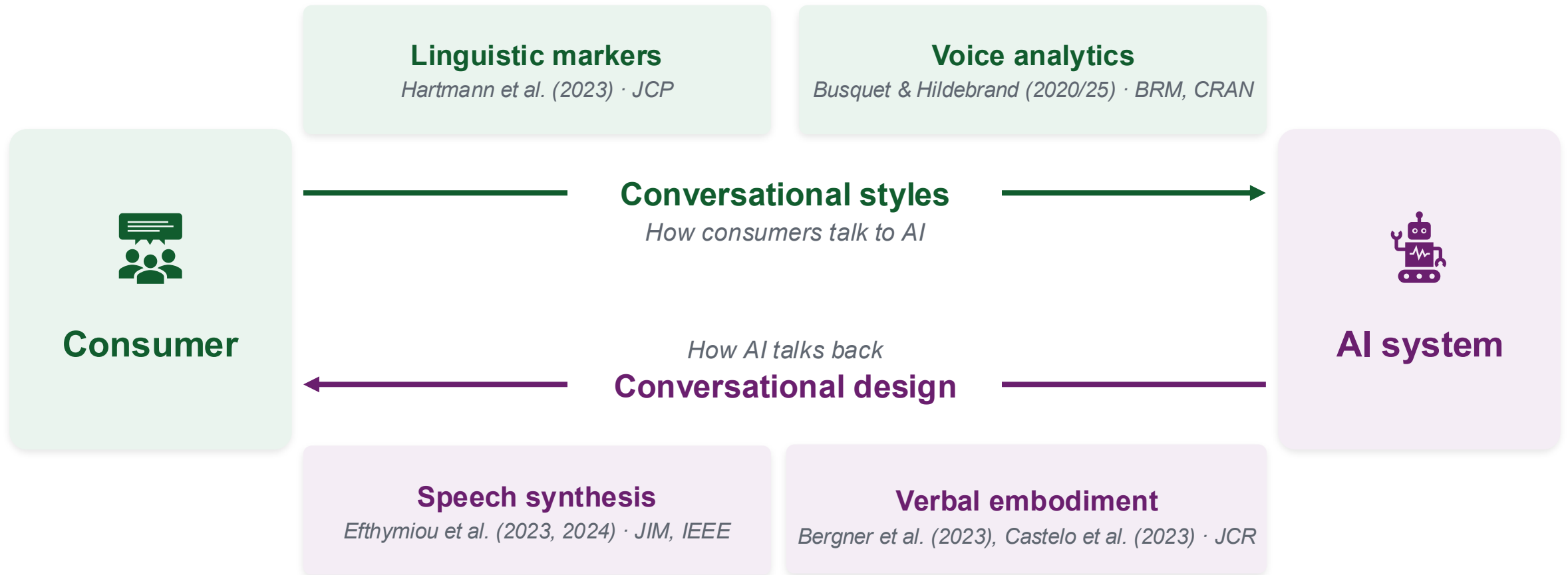


Behavioral Analytics Tools & Meta Science (ToolX)

DICE, JM (2025)
MindMiner, JCP (2023)
Meta-Analysis AI Resistance, IJRM (2025)



My Evolving Research Stream from Conversational AI...



... to Agentic Marketplaces & Economies

Human ↔ AI

Conversational experiences

- Conversational styles & linguistic markers
- Voice analytics & speech synthesis
- Verbal embodiment, conversational design



AI as intermediary

Agents shape human choice

- Agentic shopping & preference discovery
- Preference elicitation vs. construction
- Delegation dynamics



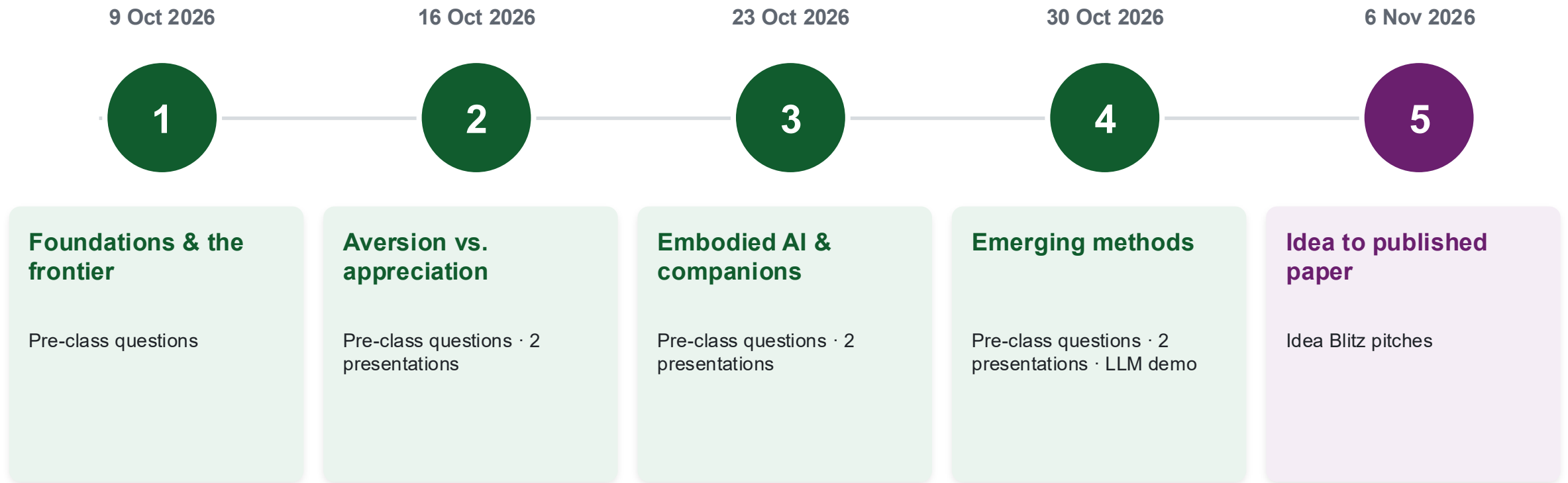
AI ↔ AI

Machines transacting with machines

- Negotiation delegation (briefed agents)
- Agent-capability asymmetry
- Market-level concentration & welfare



Course at a glance · five sessions, one research project



Running thread: each student develops one original research project, from idea and positioning to design, empirics, and target outlet. Final proposal due Fri, 13 Nov 2026.

Course objectives

OBJECTIVE 1

Map the field

An integrated map of consumer-AI research: dominant theoretical frameworks and the open questions most likely to yield top-tier publications

OBJECTIVE 2

Sharpen ideas

Identify, sharpen, and position high-contribution research ideas along the method / context / theory continuum

OBJECTIVE 3

Read critically

Critical reading and discussion of contemporary papers through structured presentations and pre-class questions

OBJECTIVE 4

Learn from leaders

See how leading scholars and former editors think about the research process via short recorded interviews

OBJECTIVE 5

Operationalize rigorously

Translate constructs into operationalizations that withstand construct, internal, external, and nomological validity concerns

OBJECTIVE 6

Plan publication

Concrete publication strategy: idea to first draft, outlet targeting, anticipating reviewer concerns

How each session works



Lecture & structured discussion

Instructor-led map of contemporary consumer-AI research and emerging methods



Paper presentations

Close reading of top-tier papers; each presenter leads the discussion



Video interviews

Short recorded interviews with leading scholars and former editors as discussion anchors

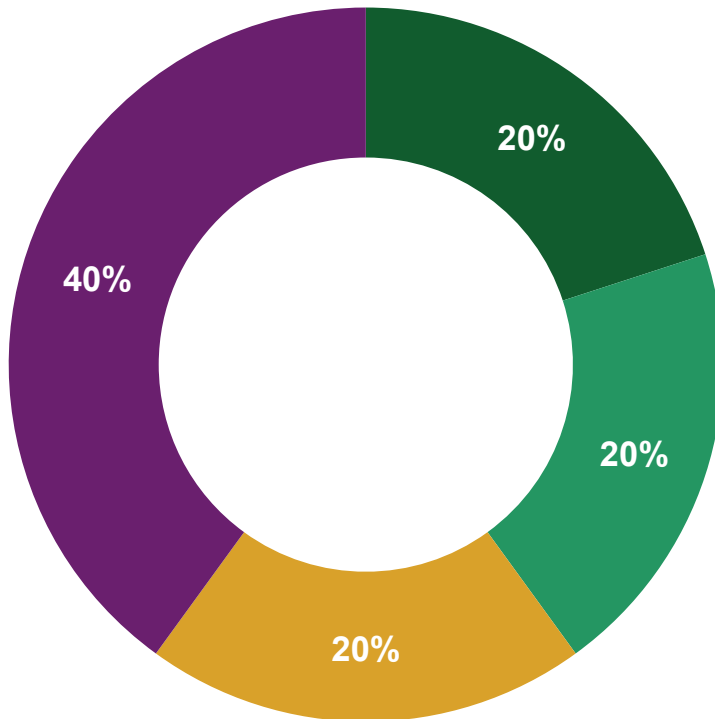


Idea development

Hands-on exercises that move your own research project forward every week

Reading-intensive: read all required papers, submit pre-class questions 24h before class, and come ready to connect each paper to your own project.

Grading



20%

Class participation & pre-class questions

Two required readings per session, questions due 24h before class

20%

Paper presentation

~10 min presentation + ~10 min student-led discussion

20%

Research-proposal "Idea Blitz"

Max. 5-minute pitch in Session 5

40%

Written research proposal

~10 double-spaced pages, due one week after Session 5

What strong participation looks like

- Questions expose a tension, mechanism, boundary condition or inference threat
- Comments build on the discussion rather than simply restating the paper
- Critique is paired with a repair: a measure, manipulation, sample or design change
- Feedback helps a peer move from an interesting topic to a testable claim

ASK

What would change your mind?

Paper presentation expectations · ~10 min + ~10 min discussion

Question & stakes

What are the authors trying to do, and why does the question matter?

Theory & design

Walk through the core framework and the design or model behind the evidence

Findings

Main results and an honest assessment of their importance

Threats

Limitations, alternative explanations, threats to inference

Extension

At least one concrete follow-up study or extension

Lead the room

Presenters lead the class discussion afterwards: bring 2–3 provocative questions

Final research proposal & Idea Blitz

- 1 Phenomenon & research question, and why it matters for consumer-AI / behavioural marketing
- 2 Prior work and your specific contribution on the theory / method / context continuum
- 3 Theoretical framework and core hypotheses
- 4 Planned studies in enough detail for a reviewer to assess the idea
- 5 Target journal and why the paper fits its scope

Written proposal · 40%

~10 double-spaced pages, 12-pt Times New Roman, 1-inch margins. Due Fri, 13 Nov 2026.

Idea Blitz · 20%

Max. 5 minutes in Session 5, following the same five-part outline.

Using AI tools in this course



Allowed — with explicit acknowledgement

- Acknowledge every functional use of a generative AI tool
- Cite AI-generated content you paraphrase, quote, or incorporate (text, images, data)
- When including AI text, state that it was AI-generated, and include the prompt where relevant
- When in doubt, ask before using it

Example acknowledgement

“I acknowledge the use of [tool, URL] to [specify the support, e.g., generate initial ideas for the study design].”

Example citation

Provider. (Year). Tool name (version). URL

Fittingly, AI is also our object of study — reflect on how you use it.

SESSION 1

Foundations & the Frontier of Consumer-AI Research

Mapping the field, its dominant frameworks, and where the next top-tier contributions are

Session 1 agenda · Fri, 9 Oct 2026 · 14:30–17:15

14:30	● Welcome, introductions & course outline	25'
14:55	● Lecture: mapping consumer-AI research	35'
15:30	● Video: Stefano Puntoni + discussion	15'
15:45	● Break	15'
16:00	● Exercise 1.1 · AI experience audit	25'
16:25	● Exercise 1.2 · Map your topic on the contribution continuum	25'
16:50	● Discussion	15'
17:05	● Wrap-up & presentation assignments	10'

● Lecture ● Discussion ● Exercise ● Video ● Presentations ● Break

Session 1 learning goals

1

Map the field

Describe consumer-AI research through an experiential lens and a strategic (mechanical / thinking / feeling) lens

2

Spot the frontier

Explain how AI can constrain, not just enable, human experience, and why that opens new questions

3

Position ideas

Locate a research idea on the method / context / theory contribution continuum

Consumers and AI: four experiences (Puntoni et al. 2021)

EXPERIENCE 1

Data capture

AI listens and collects.

Benefit:
personalization.

Tension: consumers
may feel exploited.

EXPERIENCE 2

Classification

AI sorts consumers
into types.

Benefit: relevance.

Tension: consumers
may feel
misunderstood.

EXPERIENCE 3

Delegation

AI performs tasks for
consumers.

Benefit: convenience.

Tension: consumers
may feel replaced.

EXPERIENCE 4

Social

AI acts as an
interaction partner.

Benefit: connection.

Tension: consumers
may feel alienated.

Each experience pairs a value proposition with a psychological and sociological cost.

Capability is not experience

Capability lens

- Accuracy
- Speed
- Scale
- Personalization
- Autonomy

Experience lens

- Agency
- Identity
- Dignity
- Privacy
- Connection

A strategic lens: three AI intelligences (Huang & Rust 2021)

STANDARDIZE

Mechanical AI

Automates repetitive, routine tasks; standardization and efficiency

PERSONALIZE

Thinking AI

Processes data to reach new conclusions or decisions; personalization

RELATIONALIZE

Feeling AI

Analyzes and responds to human emotions; relationalization

Applied across marketing research → strategy → action: where do consumer responses differ by intelligence type?

AI intelligences across the marketing cycle

HUANG & RUST (2021)



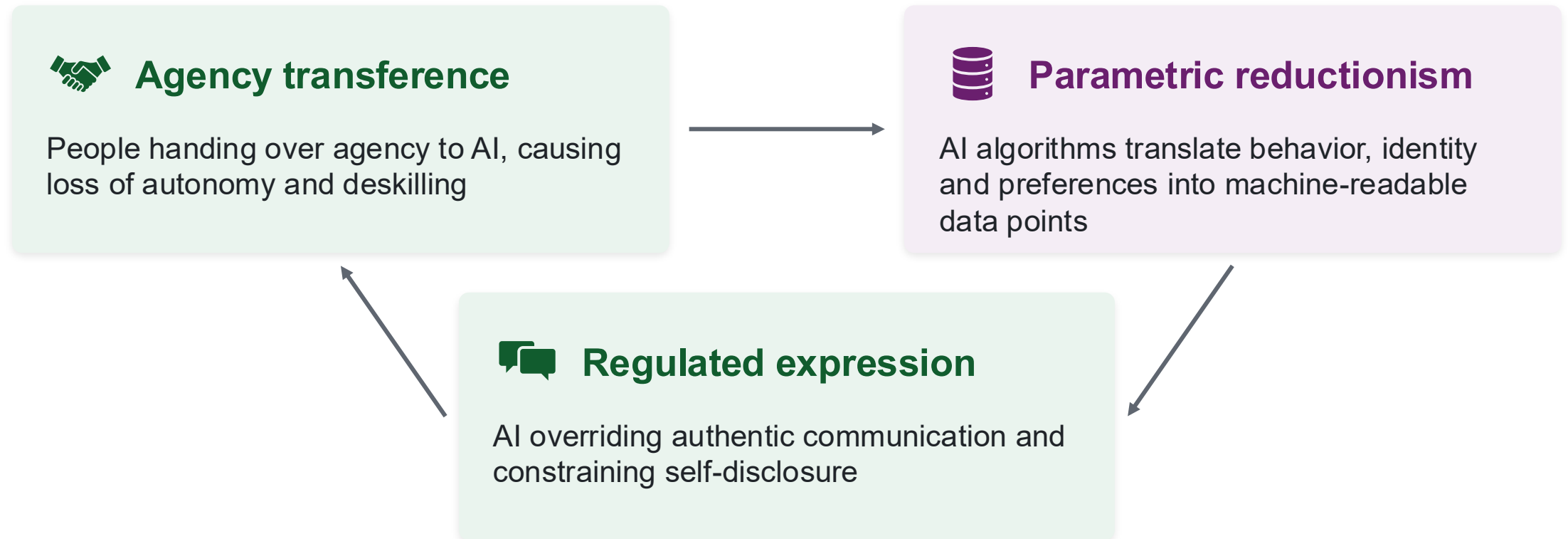
Source: Adapted from Huang, M.-H., & Rust, R. T. (2021). A strategic framework for artificial intelligence in marketing. *Journal of the Academy of Marketing Science*, 49(1), 30–50, Fig. 1. The three stages form a cycle: action feeds back into research.

Beyond adoption: how does AI constrain — not just enable — the human experience?

Valenzuela et al. (2024, JACR) · a research agenda that moves past “do consumers accept AI?”

Three mechanisms constrain the human experience

VALENZUELA ET AL. (2024)



Source: Adapted from Valenzuela, A., et al. (2024). How artificial intelligence constrains the human experience. *Journal of the Association for Consumer Research*, 9(3), Fig. 1.

Outlook: Where is your contribution? Contribution continuum

Incremental → boundary-breaking

Context

Known theory and method, applied in a new and important setting (e.g., AI companions, agentic shopping)

Method

New measurement or design that lets us see something we could not before (e.g., voice, LLM coding, field data)

Theory

A new explanation, mechanism, or boundary condition that changes how the field thinks

MacInnis et al. (2020) · Ladik & Stewart (2008): strong papers usually move on more than one dimension.

Boundary-breaking work

Stays inside a boundary

- Applies a familiar theory to a new AI setting
- Adds another outcome variable
- Documents a main effect without mechanism

Breaks a boundary

- Connects literatures that make conflicting predictions
- Shows when a valued capability becomes a constraint
- Revises who or what counts as an actor (new phenomenon, new theory, new mechanisms)

A contribution ladder

01

New context

Known effect in an
AI setting.

02

New boundary

Effect changes
under a diagnostic
condition.

03

New mechanism

Different process
explains the
behavior.

04

New construct

A concept
organizes
observations
better.

05

New paradigm

Core assumptions
about actors or
value shift.

Video interview - Stefano Puntoni



Stefano Puntoni

The experiential lens and boundary-breaking consumer-AI research

Watch for

- What does an experiential lens reveal that an adoption lens hides?
- Which consumer-AI questions does he think are still wide open?
- How does he decide a question is “big enough”?

Exercise 1.1 · AI experience audit

 25 minutes

 Pairs

Your task

1. **Your product:** pick one AI product you used this week (assistant, recommender, companion or agent).
2. **Puntoni et al.:** which of the four experiences does it create (data capture, classification, delegation, social)? Star the one with the most tension.
3. **Huang & Rust:** is its core mechanical, thinking or feeling AI?
4. **Gut check:** did it feel like a *tool* you use or a *partner* you work with?
5. **Open question:** write one research question neither framework answers.



Deliverable

One page per pair with:

- Product and AI type
- Experiences present, tension starred
- Tool or partner?
- One research question

Present it in 60 seconds.

Exercise 1.2 · Map your topic on the contribution continuum

 25 minutes

 Individual → trios

Your task

1. Write your current project idea in one sentence
2. Mark where it sits on context / method / theory
3. Trio feedback: what single change would move it one step toward theory?
4. Note the likely target journal and why



Deliverable

One sentence + continuum placement + target journal, saved as version 1 of your project log.

Session 1 takeaway

A strong consumer–AI project often starts where capability and experience pull in different directions.

Next week: explain when people reject or embrace algorithmic judgment.

Wrap-up & before next session

Today's take-aways

- Two lenses: consumer experience (Puntoni) and firm strategy (Huang & Rust)
- The frontier is moving from adoption to how AI reshapes experience
- Contribution = position on context / method / theory



To do

- Sign up for paper presentations (Sessions 2–4)
- Read the Session 2 required readings
- Pre-class questions (two readings) due Thu, 15 Oct, 14:30
- Refine your one-sentence project idea

Session 1 readings

Required (R)

Puntoni, Reczek, Giesler & Botti (2021). Consumers and AI: An Experiential Perspective. JM

MacInnis et al. (2020). Creating Boundary-Breaking, Marketing-Relevant Consumer Research. JM

Huang & Rust (2021). A Strategic Framework for AI in Marketing. JAMS

Valenzuela et al. (2024). How Artificial Intelligence Constrains the Human Experience. JACR

Optional / recommended (Rec.)

Davenport, Guha, Grewal & Bressgott (2020). How AI Will Change the Future of Marketing. JAMS

Hildebrand (2026). Consumer–AI Interactions. Oxford Bibliographies (integrative reference)

An aerial photograph of a city, likely Hong Kong, featuring a large body of water, dense urban development, and mountains in the background. The image is overlaid with a semi-transparent purple filter.

SESSION 2

Prediction Algorithms, AI Assistants & the Aversion / Appreciation Debate

When do consumers resist algorithms — and is it the label or the output?

Session 2 agenda · Fri, 16 Oct 2026 · 14:30–17:15

14:30	● Recap	10'
14:40	● Lecture: integrating aversion and appreciation	30'
15:10	● Student paper presentations (2 × 20 min)	40'
15:50	● Break	15'
16:05	● Video: Noah Castelo	10'
16:15	● Exercises	30'
16:45	● Discussion	20'
17:05	● Wrap-up	10'
● Lecture ● Discussion ● Exercise ● Video ● Presentations ● Break		

Session 2 learning goals

1

Reconcile the evidence

Explain why algorithm aversion and algorithm appreciation can both be true

2

Name the moderators

Use task type, perceived adaptivity, labels vs. outputs, and market / method / time factors to predict resistance

3

Design a clean test

Separate reactions to an AI label from reactions to AI output in an experimental design

Two literatures, one puzzle

Algorithm aversion

Dietvorst, Simmons & Massey (2015)

After seeing an algorithm err, people avoid it, even when it outperforms a human forecaster.

Algorithm appreciation

Logg, Minson & Moore (2019)

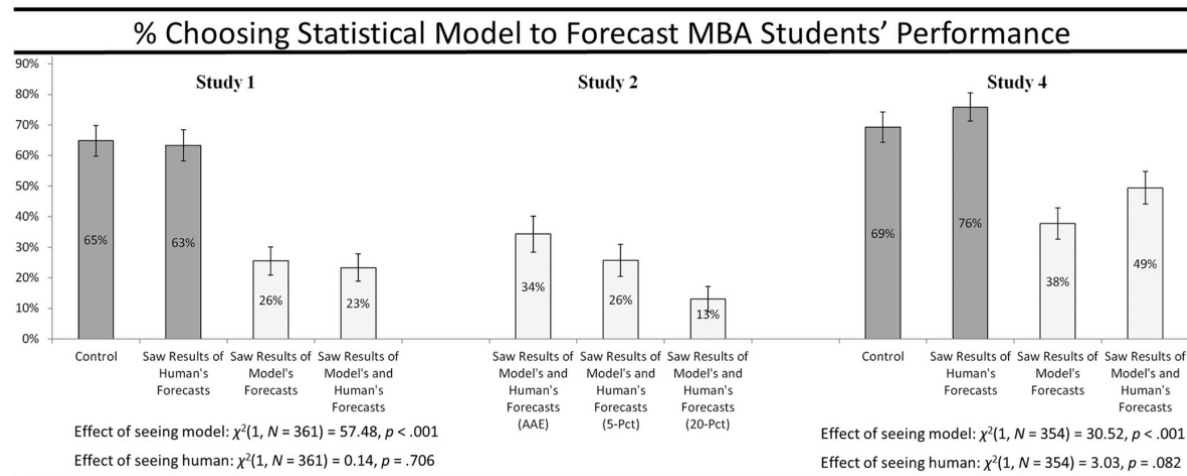
Lay people often weight advice more when it comes from an algorithm than from a person.

So when do consumers resist AI, and is it really the algorithm they resist?

Two literatures, one puzzle

Algorithm aversion

Dietvorst, Simmons & Massey (2015)



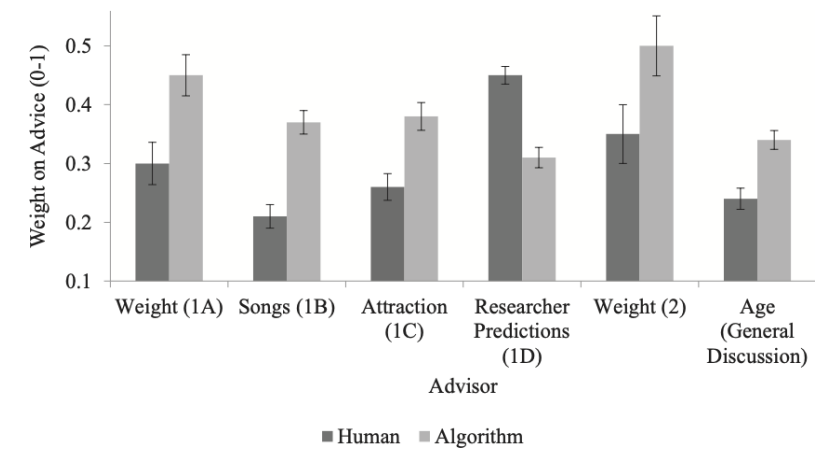
Seeing the model err cut its use from 65% to 26% (Study 1).

So when do consumers resist AI, and is it really the algorithm they resist?

Dietvorst, B. J., Simmons, J. P., & Massey, C. (2015). Algorithm aversion: People erroneously avoid algorithms after seeing them err. *JEP: General*, 144(1), 114–126.
Logg, J. M., Minson, J. A., & Moore, D. A. (2019). Algorithm appreciation: People prefer algorithmic to human judgment. *OBHDP*, 151, 90–103, Fig. 1.

Algorithm appreciation

Logg, Minson & Moore (2019)



Without seeing it err, people gave algorithmic advice more weight in 5 of 6 tests.

How aversion develops

DIETVORST ET AL.

01

Observe

A person sees
algorithmic and
human performance.

02

Notice error

An algorithm makes a
visible mistake.

03

Generalize

The error is treated as
diagnostic of the
system.

04

Avoid

The person switches
to human judgment,
even when worse.

What moves resistance up or down?

TASK

Task type

Resistance is higher for tasks seen as subjective; framing tasks as objective can reduce it (Castelo et al. 2019)

PERCEPTION

Adaptivity

Whether an algorithm is seen as learning and adapting shapes reliance (Clegg et al. 2024)

DISCLOSURE

Label vs. output

Consumers can dislike the AI label yet like the AI's output (Mariadassou et al. 2024)

SELF

Uniqueness

Consumers think AI cannot account for their unique case, e.g., in medicine (Longoni et al. 2019)

FRAMING

Preference for humans

Part of “aversion” may be a positive preference for human agents (Morewedge 2022)

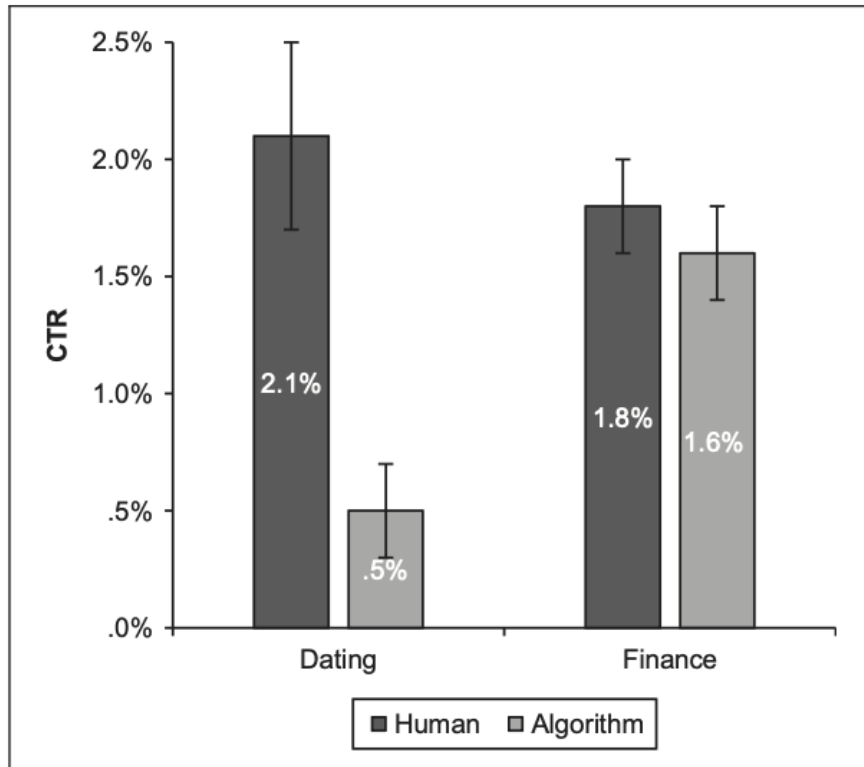
META-ANALYSIS

Markets, methods, time

Meta-analytic evidence that resistance varies systematically across these (Zehnle et al. 2025)

Consumers avoid algorithms for subjective tasks, not objective ones

CASTELO, BOS & LEHMANN (2019)



Same advice, different task

- **Dating (subjective):** ads credited to an algorithm drew a 0.5% click-through rate, against 2.1% for a human.
- **Finance (objective):** the gap almost closes, 1.6% against 1.8%.
- **Aversion tracks how subjective the task seems,** not the algorithm itself.

Error bars show standard errors. Source: Castelo, N., Bos, M. W., & Lehmann, D. R. (2019). Task-dependent algorithm aversion. *Journal of Marketing Research*, 56(5), 809–825, Fig. 2.

Adaptivity can help or hurt

Signals competence

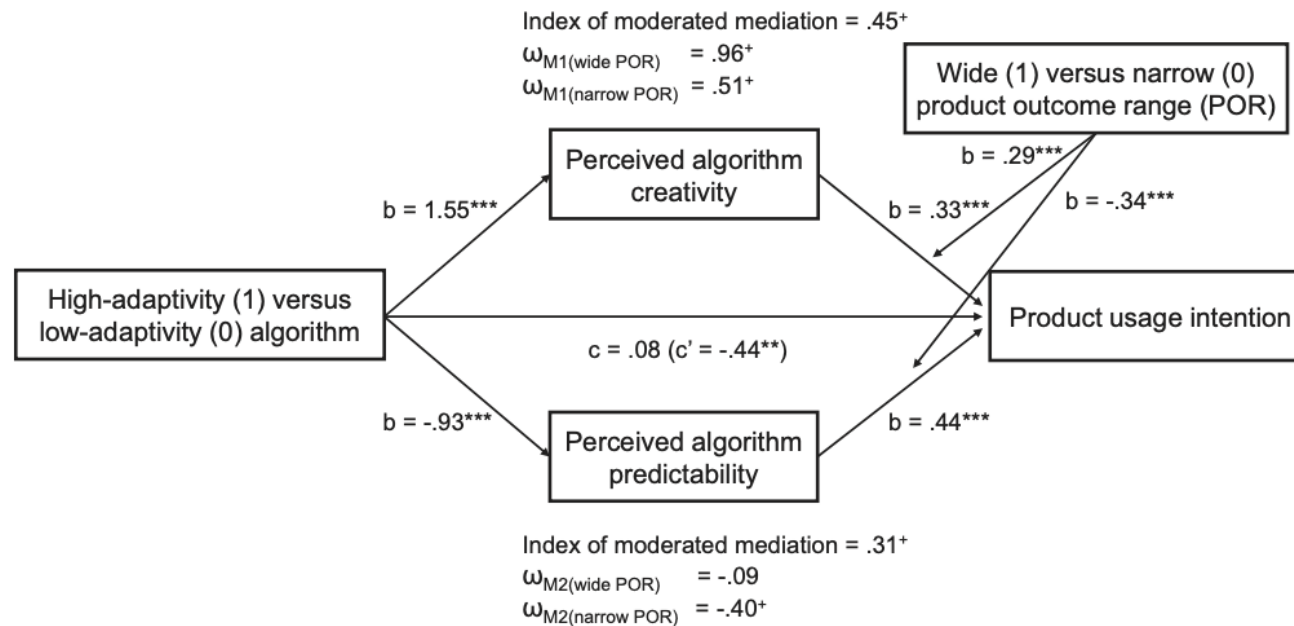
- Learning from feedback
- Personalization
- Context sensitivity
- Improving performance

Signals instability

- Unpredictable behavior
- Moving decision rule
- Reduced control
- Harder accountability

Adaptive algorithms seem more creative but less predictable

CLEGG, HOFSTETTER, DE BELLIS & SCHMITT (2024)



Two paths, one moderator

- **Creativity:** adaptivity raises perceived creativity (b = 1.55), which raises usage intention.
- **Predictability:** adaptivity lowers perceived predictability (b = -.93), which lowers usage intention.
- **Outcome range:** creativity matters more when outcomes vary widely, predictability when they are narrow.

Note: *p < .05, **p < .01, ***p < .001; ⁺significant at 5% (95% bootstrapped CIs, 10,000 draws); direct effect c' in parentheses; POR = product outcome range.

Source: Clegg, M., Hofstetter, R., de Bellis, E., & Schmitt, B. H. (2024). Unveiling the mind of the machine. *Journal of Consumer Research*, 51(2), 342–361, Fig. 5.

What our own meta-analysis asks us to separate

ZEHNLE ET AL. (2025)

Markets

High stakes, identity relevance, regulation and norms.

Methods

Hypothetical choice, real behavior, advice or delegation.

Time

Familiarity and baseline expectations change.

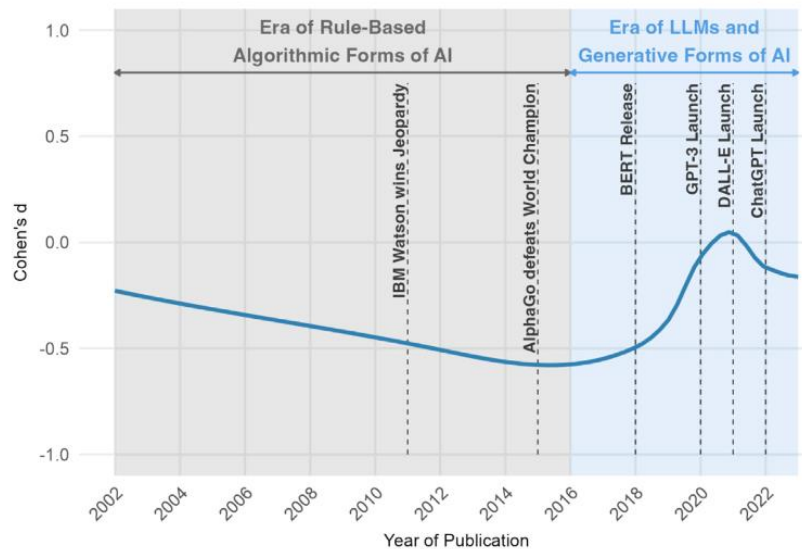
AI type

Prediction, generation, embodiment and autonomy differ.

AI resistance has faded since generative AI arrived

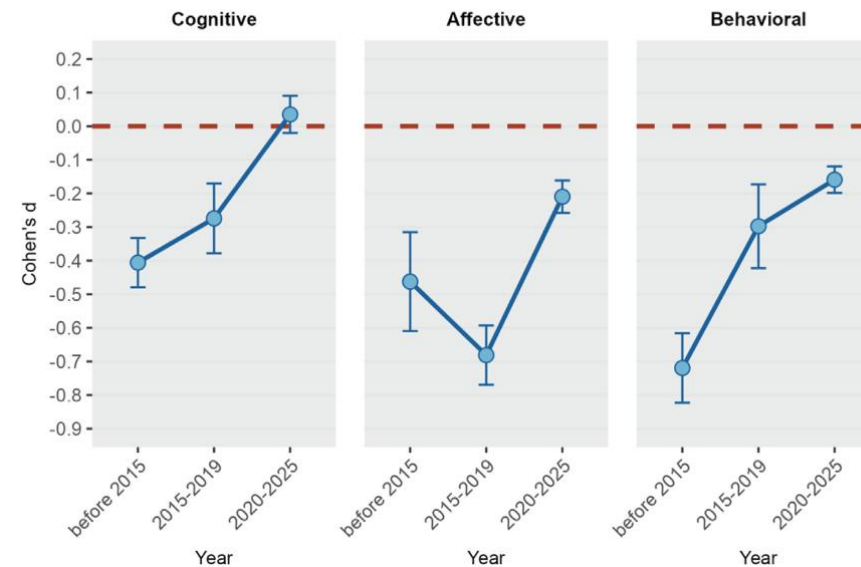
ZEHNLE ET AL. (2025)

Effects over time, 2002–2023



From $d \approx -0.5$, effects rose toward zero after 2016.

By response type and AI phase



Cognitive effects reached zero; the other two did not.

Left: Fig. 7, smoothed effect sizes with AI milestones. Right: Fig. 8, effect sizes by consumer response, before 2015, 2015–2019 and 2020–2025.

Source: Zehnle, M., Hildebrand, C., & Valenzuela, A. (2025). Not all AI is created equal: A meta-analysis revealing drivers of AI resistance across markets, methods, and time. *IJRM*.

Student presentations · Session 2

Presentation 1

Paper: _____

Presenter: _____

Presentation 2

Paper: _____

Presenter: _____

Checklist for presenters and audience

A Question & stakes · B Theory & design · C Findings · D Threats to inference · E Concrete extension

Audience: note one alternative explanation per paper. We use these in the exercise.

Video interview - Noah Castelo



Noah Castelo

Short clip on the aversion-versus-preference debate

Watch for

- What would convince them that aversion is “real”?
- Which boundary conditions do they consider settled vs. open?
- What measures do they think the field over- or under-uses?

Exercise 2.1 · Paper idea sprint

 20 minutes

 Groups of 3

Your task

1. **Start:** build on a paper from Sessions 1–2 (e.g., Puntoni, Castelo, Dietvorst) or bring your own AI phenomenon.
2. **Gap:** in one sentence, what does that work leave open?
3. **Design:** one study, one manipulation, two or three conditions.
4. **Predict:** one DV and one process measure; sketch the pattern.
5. **Extend:** one moderator that should weaken or flip the effect.
6. **Threat:** name the main confound and how you rule it out.



Deliverable

One slide per group:

- Starting point and gap
- Design with predicted pattern
- Moderator and main threat

Pitch: “We show that ____ when ____, because ____.”

Two groups present, 3 min each.

Exercise 2.2 · Boundary condition

 10 minutes

Make your central prediction conditional.

1. Expand your design and name a theoretically relevant moderator and why it changes your process
2. Draw the predicted interaction
3. Specify the most diagnostic comparison



Deliverable

A conditional hypothesis and one decisive contrast

Wrap-up & before next session

Today's take-aways

- Aversion and appreciation are conditional, not contradictory
- Moderators: task, adaptivity, label vs. output, uniqueness, markets / methods / time
- Clean designs separate the source label from output quality



To do

- Read the Session 3 required readings
- Pre-class questions due Thu, 22 Oct, 14:30
- Project log v2: add one moderator your idea depends on

Session 2 takeaway

AI response is not a stable preference. It is a judgment shaped by task meaning, comparison, error and control.

Next week: what changes when AI gains a body, a social role or a relationship with the user?

Session 2 readings

Required (R)

Dietvorst, Simmons & Massey (2015). Algorithm Aversion. JEP: General

Logg, Minson & Moore (2019). Algorithm Appreciation. OBHDP

Castelo, Bos & Lehmann (2019). Task-Dependent Algorithm Aversion. JMR

Clegg, Hofstetter, De Bellis & Schmitt (2024). Unveiling the Mind of the Machine. JCR

Mariadassou, Klesse & Boegershausen (2024). Averse to What? Current Opinion in Psychology

Zehnle, Hildebrand & Valenzuela (2025). Not All AI Is Created Equal: A Meta-Analysis. IJRM

Optional / recommended (Rec.)

Morewedge (2022). Preference for Human, Not Algorithm Aversion. Trends in Cognitive Sciences

Longoni, Bonezzi & Morewedge (2019). Resistance to Medical AI. JCR

An aerial photograph of a city, likely Hong Kong, featuring a large body of water, dense urban development, and mountains in the background. The image is overlaid with a semi-transparent purple filter.

SESSION 3

Embodied AI, Autonomous Agents & AI Companions

Voices, bodies, and relationships: when machines become social actors

Session 3 agenda · Fri, 23 Oct 2026 · 14:30–17:15

14:30	●	Recap & warm-up	10'
14:40	●	Lecture: embodiment, anthropomorphism & companions	30'
15:10	●	Student paper presentations (2 × 20 min)	40'
15:50	●	Break	15'
16:05	●	Video: Julian De Freitas + discussion	15'
16:20	●	Exercise 3.1 · Embodiment design matrix	25'
16:45	●	Exercise 3.2 · Designing a safety audit study	20'
17:05	●	Wrap-up	10'

● Lecture ● Discussion ● Exercise ● Video ● Presentations ● Break

Session 3 learning goals

1

Distinguish embodiment

Separate verbal, visual, and physical embodiment and their consequences for consumers

2

Anticipate backfire

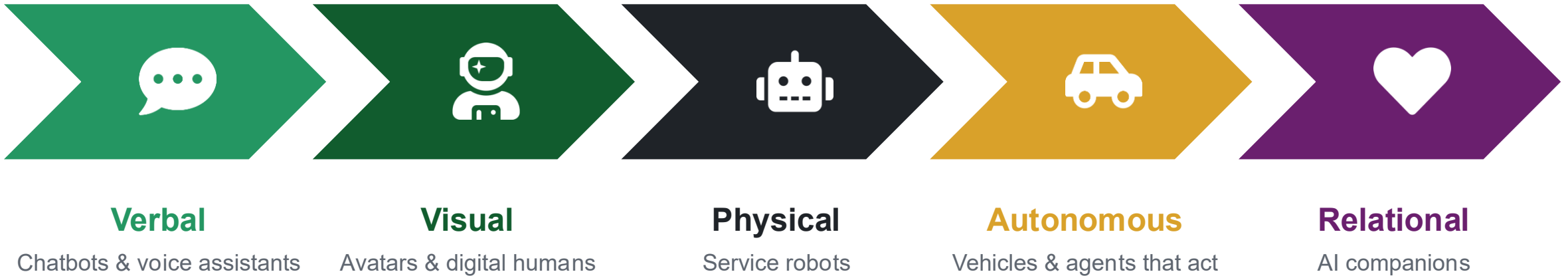
Explain when human-likeness helps and when it backfires (anger, discomfort, compensation)

3

Study safety

Design research on companion AI and wellbeing that is both rigorous and ethical

A spectrum of embodiment



Key questions: What does the consumer attribute to the agent (mind, intent, care)? What changes when the agent has a voice, a body, autonomy, or a relationship history?

Selected findings to build on

JMR

Service robots rising

Humanoid robots can trigger discomfort and compensatory consumer responses (Mende et al. 2019)

JM

Blame the bot

Anthropomorphic chatbots can backfire with angry customers (Crolic et al. 2022)

JCR

Machine talk

Verbal embodiment in conversational AI shapes consumer-brand relationships (Bergner et al. 2023)

JCR

Service bots

Why consumers react worse to service bots, and interventions that help (Castelo et al. 2023)

JM

AI coaches

Gains from AI coaching differ by agent skill level (Luo et al. 2021)

JCP

Companion safety

Generative companion apps and handling of mental-health crises (De Freitas et al. 2024)

Service bots: value or cost cutting?

Value creation cues

- Faster access
- Consistency
- Low social judgment
- Personalization

KEY READINGS

Wirtz et al. (2018), J. Service Management
Huang & Rust (2018), J. Service Research
Holthöwer & van Doorn (2023), JAMS

Cost-cutting cues

- Reduced human care
- Offloaded effort
- Lower recourse
- Firm self-interest

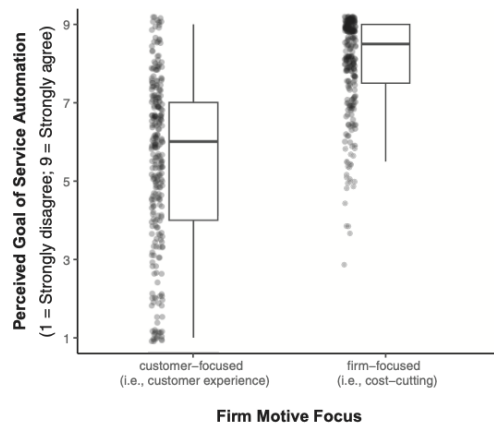
KEY READINGS

Castelo et al. (2023), J. Consumer Research
Crolic et al. (2022), J. Marketing
Luo et al. (2019), Marketing Science

Service bots: value or cost cutting?

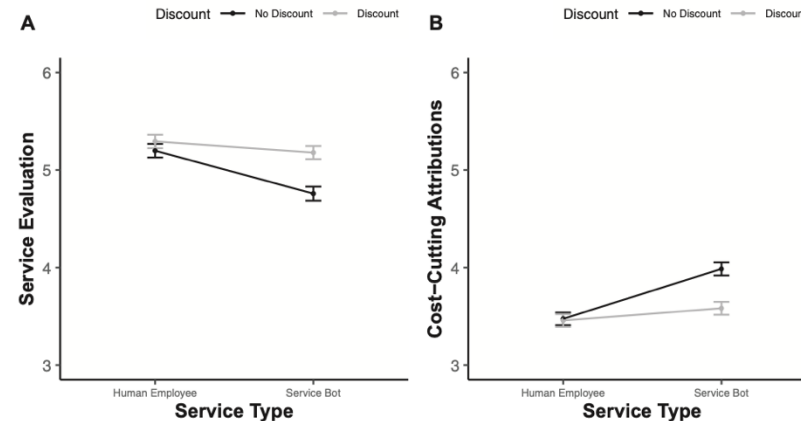
CASTELO, BOEGERSHAUSEN, HILDEBRAND & HENKEL (2023)

Main effect



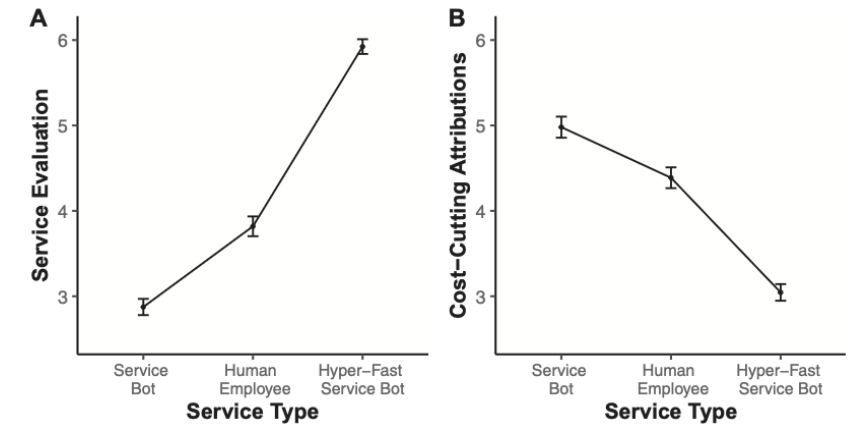
Consumers see cost cutting as the reason firms automate.

Boundary 1: discounts



Passing savings on as a discount closes the bot's evaluation gap and lowers cost-cutting attributions.

Boundary 2: hyper-fast bots



A hyper-fast bot is rated above a human employee and draws the fewest cost-cutting attributions.

Fig. 1 (left): perceived motive for service automation, 1–9 scale; boxes show median and IQR. Figs. 2 and 3: means $\pm$ SE; panel A = service evaluation, B = cost-cutting attributions.

Source: Castelo, N., Boegershausen, J., Hildebrand, C., & Henkel, A. P. (2023). Understanding and improving consumer reactions to service bots. *Journal of Consumer Research*, 50(4), 848–863.

Companions: benefit and risk

Potential benefits

- Availability
- Nonjudgmental disclosure
- Practice and rehearsal
- Perceived support

KEY READINGS

Lucas et al. (2014), Computers in Human Behavior
Maples et al. (2024), npj Mental Health Research
De Freitas et al. (2025), J. Consumer Research

Potential risks

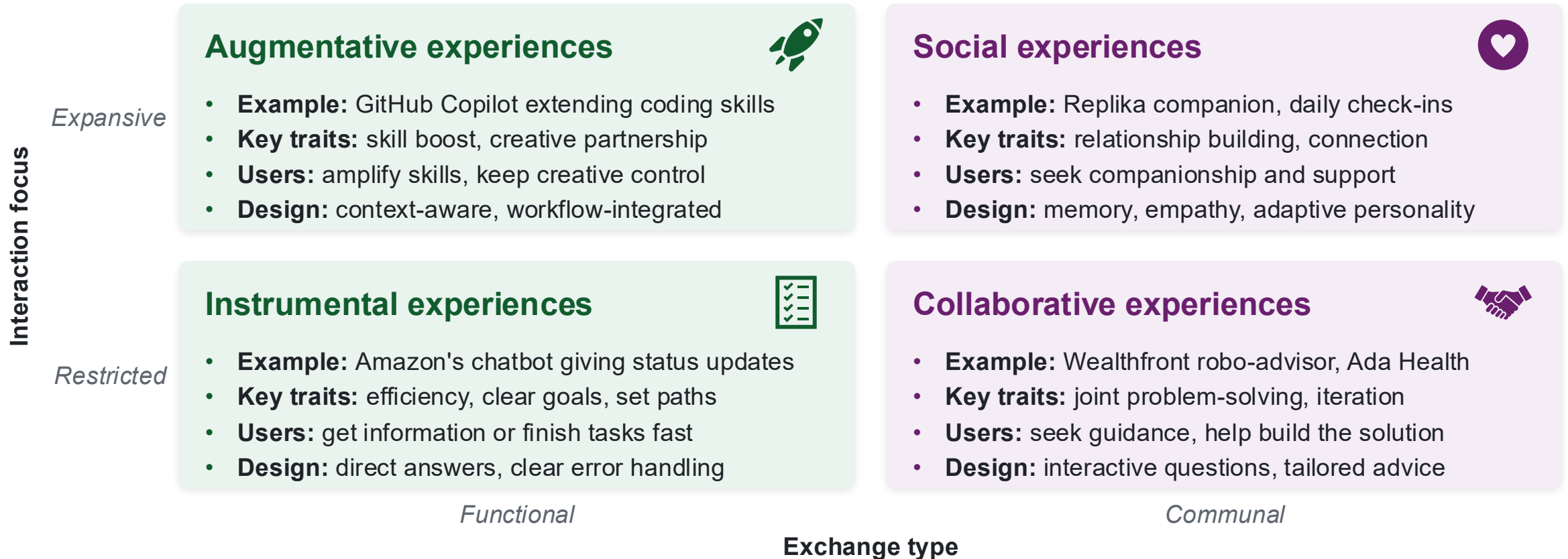
- Emotional dependence
- Manipulation and monetization
- Unsafe advice
- Displacement of human care

KEY READINGS

Laestadius et al. (2022), New Media & Society
De Freitas et al. (2024), J. Consumer Psychology
De Freitas et al. (2025), HBS Working Paper

Bot experiences differ by exchange type and interaction focus

HILDEBRAND & BERGNER (2026)



Source: Hildebrand, C., & Bergner, A. (2026). From tasks to partnerships: A typology of consumer–bot experiences. In *Handbook of Social Psychology and Consumer Behavior* (ch. 10, pp. 130–145). Edward Elgar. Adapted from Fig. 1.

Human-likeness is not a dial you simply turn up.

The same cue that builds warmth can raise expectations, anger, or discomfort.

Student presentations · Session 3

Presentation 1

Paper: _____

Presenter: _____

Presentation 2

Paper: _____

Presenter: _____

Audience focus this week

How is embodiment manipulated, and does the manipulation change only embodiment? List one confound per paper.

Student paper discussions

10 + 10 MINUTES PER PAPER

- Where does embodiment enter the causal model?
- Is anthropomorphism measured, manipulated or assumed?
- What safety outcome is missing?

Video interview · Julian De Freitas



Julian De Freitas

Companion AI safety and the ethical frontier of generative agents

Watch for

- How does he move from a societal worry to a testable question?
- What counts as evidence of harm, or safety, here?
- Where do marketing and public policy meet?

Exercise 3.1 · Embodiment design matrix

 25 minutes

 Groups of 3

Your task

1. Choose a service context (e.g., hotel check-in, bank complaint, elder care)
2. Cross embodiment (verbal / visual / physical) with consumer state (calm / angry / embarrassed)
3. Predict which cell backfires and why: name the mechanism
4. Pick the one cell you would test first and justify it



Deliverable

A 3×3 matrix with predictions and the first study you would run.

Exercise 3.2 · Ethical red team

 15 minutes

 Same groups

Try to break the design you chose in Exercise 3.1

1. Identify the most vulnerable plausible consumer / user
2. Create one misuse or failure scenario
3. Add a guardrail and a human escalation rule
4. List the ethical safeguards your IRB may ask for



Deliverable

One failure chain plus a testable guardrail.

Wrap-up & before next session

Today's take-aways

- Embodiment is multi-dimensional: verbal, visual, physical, autonomous, relational
- Human-likeness has costs as well as benefits
- Safety and wellbeing are now core marketing questions



To do

- Read the Session 4 required readings
- Pre-class questions due Thu, 29 Oct, 14:30
- Bring a laptop for the LLM text-analysis demo
- Project log v3: draft your key manipulation

Session 3 readings

Required (R)

Bergner, Hildebrand & Häubl (2023). Machine Talk. JCR

Castelo, Boegershausen, Hildebrand & Henkel (2023). Consumer Reactions to Service Bots. JCR

Crolic, Thomaz, Hadi & Stephen (2022). Blame the Bot. JM

Luo, Qin, Fang & Qu (2021). AI Coaches for Sales Agents. JM

Mende, Scott, van Doorn, Grewal & Shanks (2019). Service Robots Rising. JMR

De Freitas, Uğuralp, Oğuz-Uğuralp & Puntoni (2024). Chatbots and Mental Health. JCP

Optional / recommended (Rec.)

Holthöwer & van Doorn (2023). Robots Do Not Judge. JAMS

Schmitt (2020). Speciesism: An Obstacle to AI and Robot Adoption. Marketing Letters

An aerial photograph of a city, likely Hong Kong, featuring a large body of water, dense urban development, and mountains in the background. The image is overlaid with a semi-transparent purple filter.

SESSION 4

Emerging Methods: Open Science, Multimodal Data & LLMs as Research Tools

Credibility, new data modalities, and generative AI as both phenomenon and method

Session 4 agenda · Fri, 30 Oct 2026 · 14:30–16:55

14:30	● Recap & warm-up	10'
14:40	● Lecture: credibility & multimodal data	25'
15:05	● Student paper presentations (2 × 20 min)	40'
15:45	● Break	15'
16:00	● Video: Jochen Hartmann	10'
16:10	● Exercise 4.1 · Measurement validity	15'
16:25	● Exercise 4.2 · Silicon-sample stress test	20'
16:45	● Wrap-up	10'

● Lecture ● Discussion ● Exercise ● Video ● Presentations ● Break

Session 4 learning goals

1

Build credibility

Use preregistration, robustness checks, and transparency to protect your inferences

2

Use richer data

Choose among text, voice, image, and biometric data for a given construct

3

Use LLMs wisely

Apply LLMs as coders and simulators, and know how to validate them

Three shifts in how we do behavioural research

OPEN SCIENCE

Credibility

Replicability, robustness, reproducibility (Nosek et al. 2022); meta-analyses vs. preregistered replications (Kvarven et al. 2020); researcher degrees of freedom (Simmons et al. 2011)

NEW SIGNALS

Multimodal data

Text as a window on producers and receivers (Berger et al. 2020); automated text analysis (Humphreys & Wang 2018); voice analytics (Hildebrand et al. 2020)

GENERATIVE AI

LLMs as tools

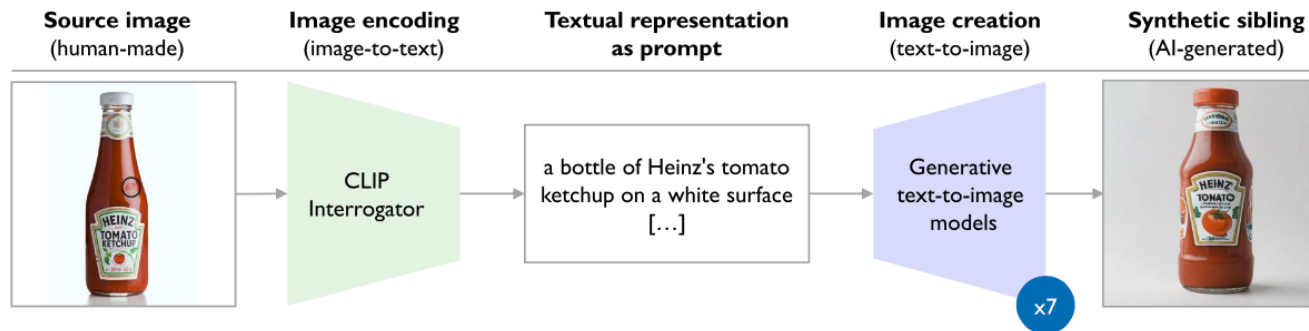
Silicon samples (Sarstedt et al. 2024); GPT for market research (Brand, Israeli & Ngwe); generative content (Hartmann et al. 2025)

Generative AI is both a research phenomenon and a research method.

Generative AI creates study stimuli at a fraction of the cost

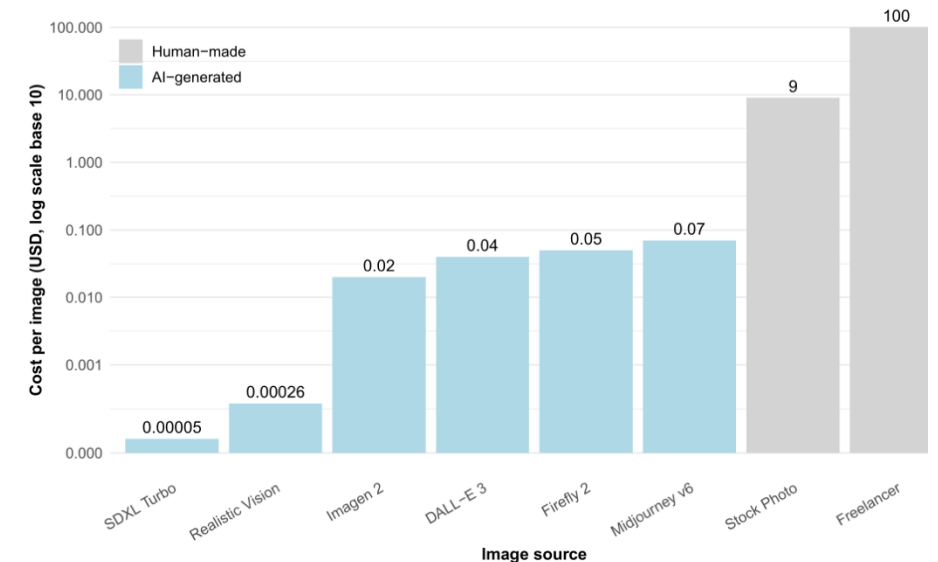
HARTMANN, EXNER & DOMDEY (2025)

One real image, seven AI derivatives



- **Any product photo** becomes a text prompt for seven text-to-image models.
- **Each original gets seven matched AI versions** to test against it.

Cost per image



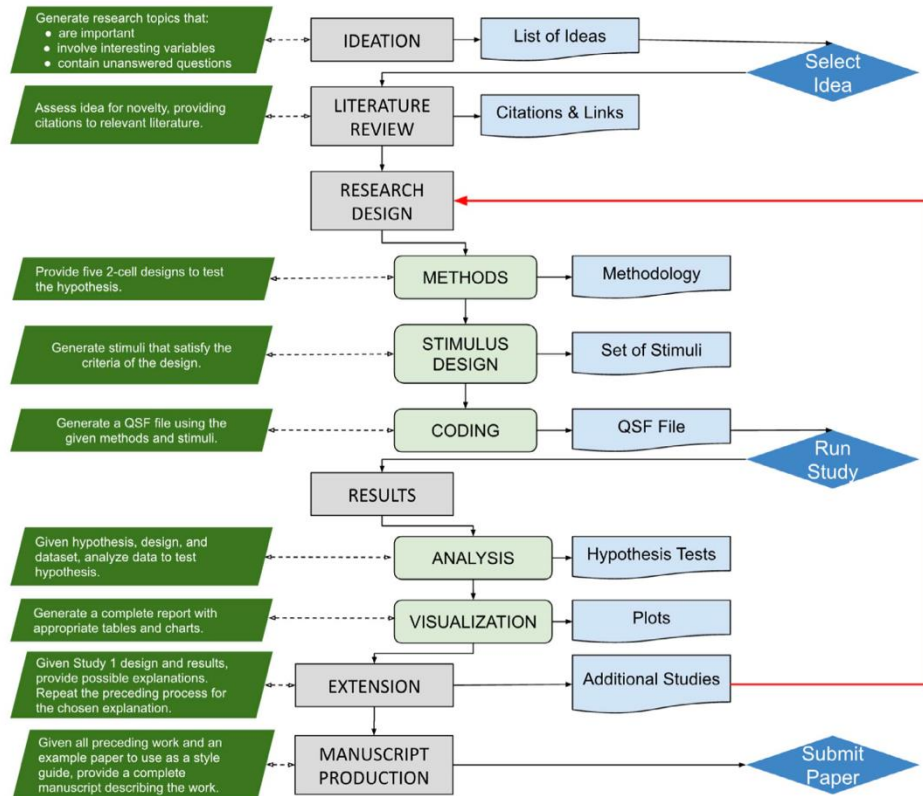
Under \$0.10 per AI image, against \$9 for stock and \$100 for a freelancer.

Left: Fig. 1, image generation procedure in studies 1 and 3. Right: Fig. 6, cost per image in USD (log scale) at creation, Q1 2024.

Source: Hartmann, J., Exner, Y., & Domdey, S. (2025). The power of generative marketing: Can generative AI create superhuman visual marketing content? *IJRM*, 42(1), 13–31.

AI can now take part in every stage of the research process

TOMAINO, COOKE & HOOVER (2025)



The cyborg behavioral scientist

- **GPT-4 is prompted at 9 of 11 stages**, from ideation and literature review to stimuli, survey code, analysis and the manuscript.
- **The researcher keeps three decisions:** select the idea, run the study, submit the paper.
- **Results loop back** into extension studies and a new research design.

Figure 1, flowchart of the AI-enabled research process. Green: prompts given to the AI; blue boxes: AI outputs; blue diamonds: human decisions.
Source: Tomaino, G., Cooke, A. D. J., & Hoover, J. (2025). AI and the advent of the cyborg behavioral scientist. *Journal of Consumer Psychology*.

The cyborg scientist

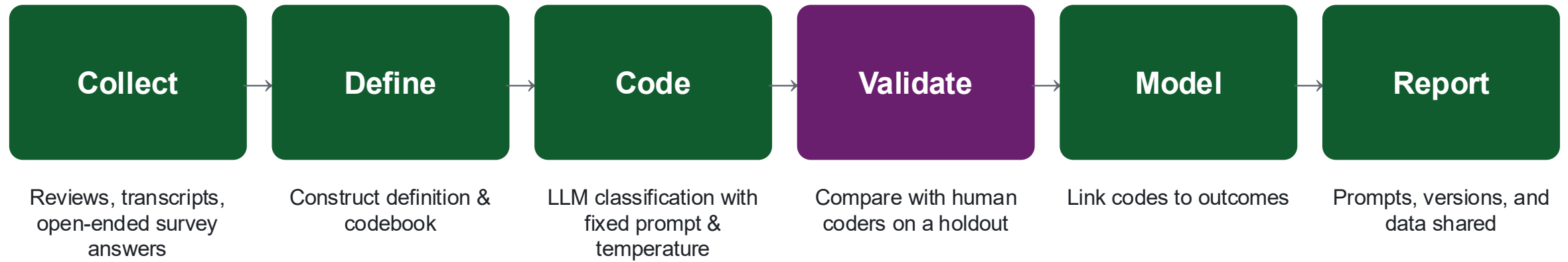
What AI can extend

- Search and synthesis
- Coding and computation
- Stimulus generation
- Adversarial critique

What cannot be delegated

- Construct judgment
- Ethical responsibility
- Inference from design
- Ownership of the scientific claim

LLM-assisted text-analysis pipeline



Validation is the step reviewers will ask about. Report agreement with human coding, prompt sensitivity, model version, and what happens when the model changes.

Text is behavioral data

**Words reveal attention, identity, emotion and strategy,
but only through a defensible measurement model.**

A classifier score is not automatically a construct.

Uniting the tribes

BERGER ET AL. (2020)

Theory-first

Start from a construct and operationalize linguistic indicators.

Data-first

Discover patterns and then establish meaning and validity.

Prediction

Optimize out-of-sample performance.

Explanation

Estimate a theoretically interpretable relationship.

Validate automated text analysis

01

Define

What is the target construct?

02

Annotate

How is ground truth created?

03

Evaluate

What errors occur, for whom?

04

Transport

Does it generalize across domains?

05

Audit

Can another researcher reproduce it?

Three credibility goals

NOSEK ET AL. (2022)

Reproducibility

The same data and code
yield the same result.

Replicability

New data and the same
design yield a compatible
result.

Robustness

Reasonable analytical
choices do not overturn
the conclusion.

An auditable LLM pipeline

01

Freeze inputs

Corpus, model and prompt.

02

Create benchmark

Human-coded validation set.

03

Run blind

Separate tuning from evaluation.

04

Inspect errors

Classes, subgroups and edge cases.

05

Report

Prompts, versions, metrics and limitations.

The pipeline is the product

Credibility depends on the full chain from construct to data to transformation to inference.

Open materials matter because hidden decisions accumulate.

LLMs for market research

BRAND ET AL. (2024)

Generate

Hypotheses, stimuli and interview probes.

Transform

Summaries, coding assistance and feature extraction.

Simulate

Synthetic responses for exploration, not automatic substitution.

Evaluate

Human benchmarks, error analysis and documented prompts.

Silicon samples

Promise

- Fast pretesting
- Rare scenario generation
- Theory stress tests
- Stimulus and scale development

Failure modes

- Model–human mismatch
- Unknown training overlap
- Homogenized responses
- Prompt and version sensitivity

Student presentations · Session 4

Presentation 1

Paper: _____

Presenter: _____

Presentation 2

Paper: _____

Presenter: _____

Audience focus this week

Which validity concern (construct, internal, external, nomological) does the method address best, and which does it leave open?

Student paper discussions

10 + 10 MINUTES PER PAPER

- Locate the most consequential researcher degree of freedom
- Identify the validation set and benchmark
- Separate predictive performance from construct validity
- Propose one reproducibility artifact the authors should share

Video interview · Jochen Hartmann



Jochen Hartmann

Generative AI as both research phenomenon and research method

Watch for

- How does he validate AI-generated stimuli and measures?
- When is “superhuman” content a finding vs. an artifact?
- What should a PhD student build skills in now?

Guest interview prompts

CONVERSATION WITH PROF. JOCHEN HARTMANN

Constructs

Which marketing constructs are easiest to mismeasure with AI?

Visuals

How should researchers validate generated stimuli?

Disclosure

What details make AI-assisted work reproducible?

Frontier

Where can AI change theory, not just efficiency?

Exercise 4.1 · Measurement validity

 15 minutes

Audit the weakest measure in your proposal.

1. State your primary construct in one sentence
2. Explain how the observed measure maps onto it
3. Name the strongest rival interpretation
4. Add one convergent or discriminant validation test



Deliverable

A measurement argument plus one validation study

Exercise 4.2 · Silicon-sample “stress test”

 20 minutes

 Pairs

Your task

1. Take a classic consumer-behaviour effect related to your project
2. Write how you would simulate it with an LLM “sample”
3. List three ways the simulation could mislead (e.g., training data, demand effects)
4. Decide: pretest tool, hypothesis generator, or replacement for humans?



Deliverable

A one-paragraph verdict with your decision and the safeguard you would report.

Wrap-up & before next session

Today's take-aways

- Credibility is designed in, not added at the end
- New data modalities need construct validation
- LLMs are powerful coders and simulators, but only with validation



To do

- Read the Session 5 required readings
- Prepare your 5-minute Idea Blitz
- Project log v4: draft methods paragraph
- Final proposal due Fri, 13 Nov 2026

Session 4 readings

Required (R)

Nosek et al. (2022). Replicability, Robustness, and Reproducibility in Psychological Science. *Annual Review of Psychology*

Sarstedt, Adler, Rau & Schmitt (2024). LLMs to Generate Silicon Samples. *Psychology & Marketing*

Hartmann, Exner & Domdey (2025). The Power of Generative Marketing. *IJRM*

Berger, Humphreys, Ludwig, Moe, Netzer & Schweidel (2020). Uniting the Tribes: Text for Marketing Insight. *JM*

Optional / recommended (Rec.)

Kvarven, Strømmland & Johannesson (2020). Meta-Analyses vs. Multi-Lab Replications. Nature Human Behaviour

Simmons, Nelson & Simonsohn (2011). False-Positive Psychology. Psychological Science

Tomaino, Cooke & Hoover (2025). AI and the Advent of the Cyborg Behavioral Scientist. JCP

Humphreys & Wang (2018). Automated Text Analysis for Consumer Research. JCR

Netzer, Lemaire & Herzstein (2019). When Words Sweat. JMR

An aerial photograph of a city, likely Hong Kong, featuring a large body of water (Victoria Harbour) in the background, surrounded by mountains. The foreground shows a dense urban area with numerous buildings and green spaces. The image is overlaid with a semi-transparent purple filter.

SESSION 5

From Idea to Published Paper

Contribution, importance, writing, and the review process

Session 5 agenda · Fri, 6 Nov 2026 · 14:30–17:15

14:30	● Recap: what we built this term	10'
14:40	● Lecture: contribution, importance & the review process	30'
15:10	● Break	15'
15:25	● Exercise 5.1 · Proposal Blitz pitches (5 min each)	60'
16:25	● Exercise 5.2 · Reviewer 2 swap and feedback	15'
16:40	● Course synthesis & proposal Q&A	25'
17:05	● Course evaluation & close	10'

● Lecture ● Discussion ● Exercise ● Video ● Presentations ● Break

Session 5 learning goals

1

Articulate contribution

Write a contribution statement that creates and helps readers appreciate new knowledge

2

Argue importance

Explain why your question matters to scholars and practice

3

Navigate review

Anticipate the editorial and reviewer concerns that decide a manuscript's fate

What makes a paper publishable?

CONTRIBUTION

Knowledge creation & appreciation

Create new knowledge and help readers see why it matters
(Janiszewski, Labroo & Rucker 2016)

IMPORTANCE

Importance

Why important marketing issues are under-studied, and how to fix it (Kohli & Haenlein 2021; Stremersch 2021)

CRAFT

Interesting & impactful

Start from phenomena, use theory to explain them, write for clarity (Tellis 2017)

PROCESS

The review journey

Lessons from thousands of editorial decisions at JMR (Gupta et al. 2023)

What makes a paper publishable?

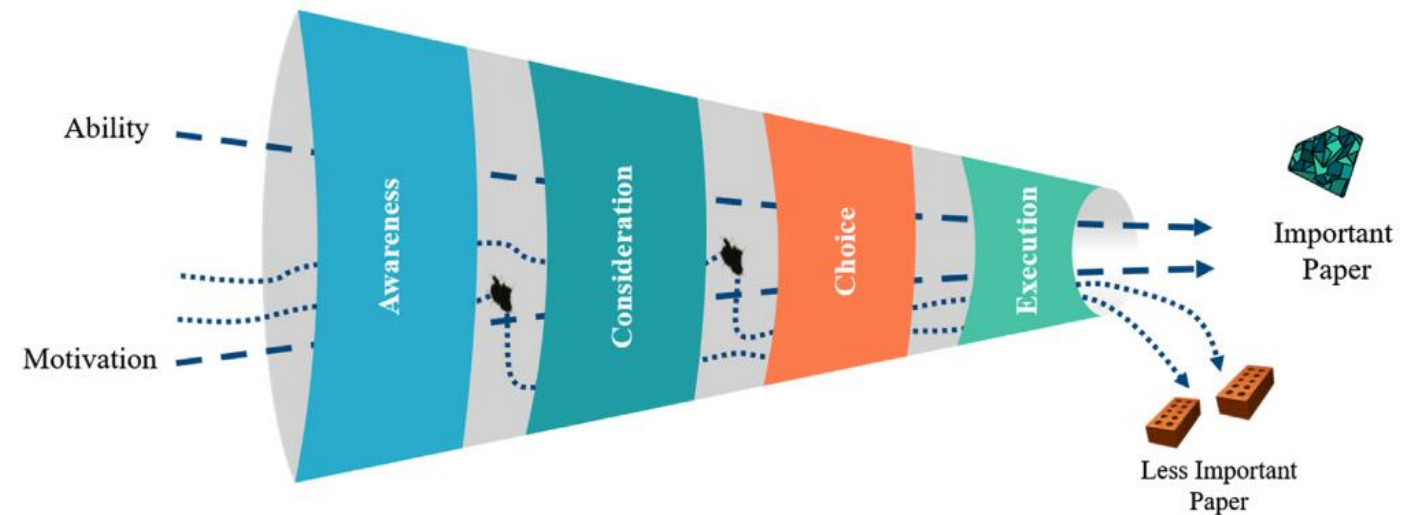
STREMERSCH (2021)

Importance and relevance

Important	Science Leadership <i>e.g., How to correct for endogeneity of managerial decisions?</i>	Thought Leadership <i>e.g., How to make an innovation portfolio more customer centric?</i>
	Puzzling Science <i>e.g., How to remain diplomatic when creating a typology?</i>	Applied science <i>e.g., How to optimize a Facebook campaign for an FMCG brand?</i>
Not so important	Not so managerially relevant	Managerially relevant

Aim for thought leadership: important and managerially relevant.

Ability and motivation along the project funnel



Projects need both ability and motivation to come out of the funnel as important papers.

Fig. 1 (left): importance and relevance. Fig. 2 (right): ability and motivation along the project funnel.
Source: Stremersch, S. (2021). *International Journal of Research in Marketing*, 38, 12–17.

From topic to contribution

01

Topic

AI companions in service.

02

Phenomenon

Users disclose more to nonhuman agents.

03

Tension

Disclosure helps now but may deepen dependence.

04

Mechanism

Reduced evaluation concern changes self-regulation.

05

Contribution

Relational safety can create longitudinal vulnerability.

Knowledge creation and appreciation

Create knowledge

- Ask an unanswered question
- Develop a mechanism
- Test rival explanations
- Specify boundaries

Enable appreciation

- Make importance vivid
- Connect to a recognized conversation
- Use a memorable organizing idea
- Show what changes if you are right

What makes an issue important?

KOHLI & HAENLEIN (2021)

Magnitude

Many people, large consequences or meaningful welfare.

Change

A market or behavior is shifting faster than theory.

Blind spot

Current concepts systematically miss the phenomenon.

Decision relevance

Consumers, firms or policymakers face a real choice.

Importance must be argued

A popular technology does not automatically create an important marketing question.

Show the decision, consequence and theoretical blind spot.

Phenomena, theory and writing

Phenomenon

Makes the reader care and anchors the puzzle.

Theory

Explains why the pattern occurs and when it changes.

Evidence

Rules out alternatives and establishes scope.

Writing

Makes the logic easy to retrieve and evaluate.

Choose an outlet by conversation

Audience

Who needs this insight?

Conversation

Which literature will cite it?

Evidence norm

What designs and scope are expected?

Promise

Can the paper deliver what its framing implies?

A five-paragraph introduction

01

Phenomenon

What is
happening?

02

Problem

Why do current
accounts fall
short?

03

Insight

What is your core
explanation?

04

Evidence

How will you
establish it?

05

Contribution

What changes in
the literature?

The five-paragraph outline

- | | | |
|----------|----------------------------------|--|
| 1 | Phenomenon & question | What is happening, and what do we not know? |
| 2 | What we know | The closest prior work, and its gap |
| 3 | Theory & hypotheses | Your mechanism and predictions |
| 4 | Evidence | Studies, data, and key design choices |
| 5 | Contribution & outlet | What changes for theory and practice, and why this journal |

Work from a study portfolio

Study 1

Establish the effect cleanly.

Study 2

Demonstrate the mechanism.

Study 3

Rule out the strongest rival.

Study 4

Show a consequential boundary or field relevance.

Threats reviewers will see (incomplete selection!)

Internal validity

- Demand and confounds
- Selective attrition
- Post-treatment measures
- Too flexible analysis

External and construct validity

- Measure–construct issues
- Artificial task meaning
- Narrow population
- Weak manipulation / realism

Exercise 5.1 · Proposal Blitz

 5 minutes

Pitch your project in five minutes. Stop when the timer stops.

1. Phenomenon and why it matters
2. Theoretical tension and core insight
3. Two decisive studies
4. Contribution and target conversation



Deliverable

A five-minute pitch plus one slide of study architecture

Exercise 5.2 · Proposal Blitz Group Discussion Follow-up

 15 minutes

 Individual → peer swap

Your task

1. Write one paragraph per box for your project
2. Swap with a neighbour who acts as Reviewer 2 (5 minutes)
3. Reviewer: mark the weakest paragraph and the most likely rejection reason



Deliverable

A complete five-paragraph outline:
the skeleton of your written proposal.

Peer feedback: two lenses

Reviewer lens

- What is the claim?
- Sufficient contribution (theory / method / context)?
- Which overlooked alternatives?
- What would be a fatal flaw?

Editor lens

- Why now?
- Why marketing?
- Why this journal?
- Who will care?
- What changes if true?

Final note on written proposal requirements

- Approximately 10 double-spaced pages
- Phenomenon and research question
- Literature positioning and contribution
- Conceptual framework and hypotheses
- Studies detailed enough for reviewer assessment
- Target journal and explicit fit

Use the proposal as a blueprint working toward a full paper.

Grading rubric

Importance

Consequential question and clear motivation.

Theory

Coherent mechanism, boundaries and novelty.

Evidence

Diagnostic, feasible and transparent studies.

Communication

Precise logic, strong structure and professional execution.

Course wrap-up & next steps

Today's take-aways

- Contribution = creating knowledge and helping readers appreciate it
- Importance must be argued, not assumed
- Anticipate the reviewer before the reviewer does (at least as best as we can)



To do

- Revise your proposal using today's Blitz feedback
- Written proposal (~10 pages) due Fri, 13 Nov 2026
- Acknowledge any AI use (Approach 3)
- Complete the course evaluation

Session 5 readings

Required (R)

Janiszewski, Labroo & Rucker (2016). Knowledge Creation and Knowledge Appreciation. JCR

Gupta, Danaher, Mittal & Morrin (2023). A Manuscript's Journey Through Peer Review. JMR

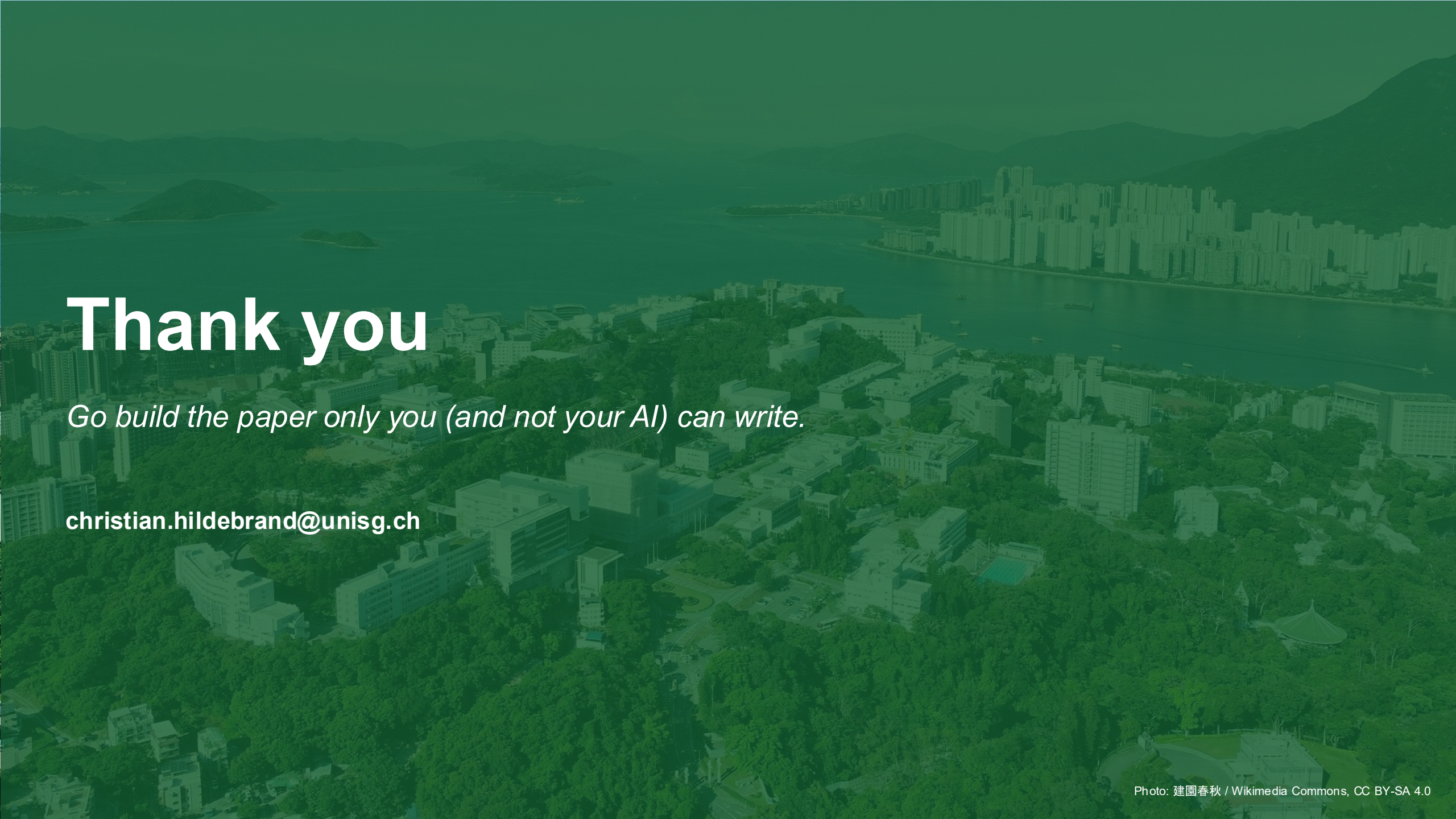
Kohli & Haenlein (2021). Factors Affecting the Study of Important Marketing Issues. IJRM

Stremersch (2021). The Study of Important Marketing Issues: Reflections. IJRM

Tellis (2017). Interesting and Impactful Research. JAMS

Optional / recommended (Rec.)

Ladik & Stewart (2008). The Contribution Continuum. JAMS



Thank you

Go build the paper only you (and not your AI) can write.

christian.hildebrand@unisg.ch